

Notes: Cortés et al. (JFE, 2020)

Stress tests and small business lending

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1 Big picture

Question. Do post-crisis supervisory stress tests change the supply of small-business credit? If so, do affected banks cut quantities, raise prices, reduce risk, or simply reallocate lending across markets?

Setting. The paper studies U.S. bank holding companies subject to Federal Reserve stress tests, especially the CCAR/DFAST regime after the financial crisis. The main years are 2012–2016 for stress-test exposure, 2012–2015 for annual CRA small-business loan quantities, and 2012Q2–2017Q2 for loan-level pricing in the Survey of Terms of Business Lending.

Core challenge. Stress-tested banks are large and different from non-stress-tested banks. A simple comparison of stress-tested and non-tested banks would mix the effect of stress testing with bank size, business model, risk appetite, and local-market differences.

Main empirical strategy. The paper mostly compares banks *within the stress-tested group*. Some stress-tested BHCs are more affected by the tests because their stressed capital ratios fall closer to regulatory minimums. The paper asks whether banks with greater stress-test exposure change small-business credit more than other stress-tested banks lending in the same local markets.

Main measure. Stress-test exposure is based on how close a BHC's stressed capital ratios come to regulatory thresholds, or how much the Tier 1 capital ratio falls under the stress scenario. The signs are chosen so that larger values always mean the BHC is more affected by the stress test.

Main quantity result. Banks more affected by stress tests reduce small-business loan originations in riskier local economies. The key result is not a uniform reduction in all lending, but a reallocation away from risky markets.

Local-knowledge result. The quantity contraction is concentrated in markets where the bank does *not* have local branches. Where banks have a local branch presence, quantities fall much less.

Main price result. Stress-test exposure raises interest rates on small-business loans. The price increase is strongest for risky loans and for loans made in markets where the bank has a local branch presence.

Mechanism. Stress tests increase the shadow capital cost of risky lending. In local markets, banks have better private information and stronger relationships, so they can charge higher rates for risky loans without losing all borrowers. In non-local markets, they have less local information and weaker relationships, so they exit or reduce risky lending instead.

Risk-composition result. Banks more affected by stress tests reduce the share of high-risk loans, especially in non-local markets. This supports the view that the quantity and price effects reflect a change in credit supply and risk taking.

Aggregate result. The paper does not find that stress tests reduce total small-business credit at the county level. Smaller local banks appear to expand their market share in counties formerly reliant on stress-tested non-local lenders.

Economic takeaway. Stress tests do not simply shrink small-business lending. They change *who lends, where lending occurs, and how risk is priced*. Large stress-tested banks reduce or reprice risky small-business credit, while smaller local banks partly fill the gap.

2 Data and measurement (key objects)

2.1 Stress-test setting

The paper studies stress tests run by the Federal Reserve after the financial crisis. The 2009 Supervisory Capital Assessment Program was followed by annual stress testing for large BHCs under CCAR and DFAST.

Stress tests project each BHC's capital ratios under forward-looking macroeconomic stress scenarios. The relevant public outputs are minimum stressed capital ratios over the nine-quarter planning horizon.

Interpretation. A BHC whose stressed capital ratio comes close to a regulatory minimum faces more pressure to reduce risk, retain capital, or adjust lending.

2.2 Stress-test exposure measures

Let h index the BHC and t the stress-test year. The paper uses three related measures.

Minimum stress-test distance. The raw distance is

$$D_{h,t}^{\min} = \min \begin{cases} \text{Stressed Tier 1 capital}_{h,t} - 6\%, \\ \text{Stressed total risk-based capital}_{h,t} - 8\%, \\ \text{Stressed leverage ratio}_{h,t} - 4\%. \end{cases} \quad (1)$$

Because a smaller distance means the bank is more constrained, the paper reverses the sign when using this as an exposure variable.

Tier 1 capital distance. The raw measure is

$$D_{h,t}^{T1} = \text{Stressed Tier 1 capital}_{h,t} - 6\%. \quad (2)$$

Again, the sign is reversed so that higher exposure means the bank is more affected.

Tier 1 capital stress-test exposure. This measure captures the decline in Tier 1 capital under stress:

$$\text{Exposure}_{h,t}^{T1} = \text{Initial Tier 1 capital}_{h,t} - \text{Stressed Tier 1 capital}_{h,t}. \quad (3)$$

Interpretation. The three measures capture related but distinct ways in which the stress test binds: closeness to regulatory thresholds and the size of projected losses under stress.

2.3 CRA small-business lending data

The main quantity data come from Community Reinvestment Act disclosure reports. CRA data record small-business loan originations by bank and county.

The paper treats loans with commitment amounts below 1 million as small-business loans. The CRA quantity sample focuses on banks owned by stress-tested BHCs and covers 2012–2015.

The main dependent variable is annual bank-county loan growth:

$$\text{Loan growth}_{i,c,t} = \frac{\text{Originations}_{i,c,t} - \text{Originations}_{i,c,t-1}}{(\text{Originations}_{i,c,t} + \text{Originations}_{i,c,t-1})/2}. \quad (4)$$

This midpoint growth rate is bounded between -2 and $+2$ and reduces problems from small denominators.

Sample restriction. The regressions require a bank to have made at least five loans in the county in the prior year. This removes very small bank-county cells while preserving almost all loan volume.

Interpretation. CRA data are useful because they show where banks originate small-business loans. Their limitation is that they do not observe borrower-level risk.

2.4 Local market risk: employment beta

Because CRA data do not contain borrower risk, the paper constructs a county-level risk measure. First, it estimates industry employment betas: the sensitivity of employment growth in industry j to aggregate employment growth.

Then it aggregates these industry betas to the county-year level:

$$\text{Employment beta}_{c,t} = \sum_j \omega_{c,j,t} \text{Employment beta}_j, \quad (5)$$

where $\omega_{c,j,t}$ is industry j 's employment share in county c at time t .

Interpretation. Counties with higher employment beta are more cyclically sensitive and are therefore riskier small-business lending markets under macroeconomic stress.

2.5 Local versus non-local lending

For the CRA quantity analysis, a bank is local in a county if it has a branch in that county. For the STBL pricing analysis, a bank is local if it has a branch in the borrower's state.

Interpretation. Branch presence proxies for local knowledge, borrower relationships, and informational advantage. The distinction is central because the paper argues that banks price risk when they have local information and reduce quantities when they do not.

2.6 STBL loan-level pricing data

The pricing data come from the Federal Reserve's Survey of Terms of Business Lending. The STBL records new business loans made by sampled banks during one business week each quarter.

For each loan, the paper observes the interest rate, loan size, maturity, collateral status, whether the rate is floating, whether the loan is syndicated, prepayment penalties, and the lender's internal risk rating.

The paper keeps loans with commitments below 1 million, excludes distressed loans, and excludes unrated loans because risk ratings are important for identification.

Risk rating. The lender's internal risk rating ranges from 1 to 4, where 1 is safest and 4 is riskiest.

Timing. Stress-test exposure is mapped to the next four quarterly STBL surveys. For example, stress-test results disclosed in March 2013 are matched to loans in May, August, and November 2013 and February 2014.

Interpretation. STBL data allow the authors to see whether stress tests raise loan prices and whether the increase is concentrated among riskier loans.

2.7 Aggregate county-level outcomes

To ask whether stress tests reduce total credit, the paper aggregates CRA loan originations to the county-year level.

It constructs county-level exposure to local and non-local stress-tested banks using pre-sample loan shares. It then asks whether counties more exposed to affected stress-tested banks experience lower total small-business lending growth or greater market-share gains by small local banks.

Interpretation. This separates bank-level supply changes from the aggregate availability of credit in a county.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline CRA loan-growth model

The baseline quantity model is

$$\text{Loan growth}_{i,c,t} = \beta_1 \text{Exposure}_{h,t} + X'_{h,t-1} \Gamma + \gamma_{c,t} + \varepsilon_{i,c,t}, \quad (6)$$

where i indexes the subsidiary bank, h the parent BHC, c the county, and t the year.

$\text{Exposure}_{h,t}$ is one of the stress-test exposure measures. $X_{h,t-1}$ includes BHC controls such as size, C&I loans/assets, nonperforming loans, ROA, deposit funding, consumer loans, mortgage loans, trading assets, and liquidity. $\gamma_{c,t}$ is a county-by-year fixed effect.

Interpretation. The county-year fixed effect compares banks lending in the same local market in the same year. This helps control for local credit demand and local economic conditions.

3.2 Stress-test exposure interacted with local-market risk

The key quantity specification is

$$\begin{aligned} \text{Loan growth}_{i,c,t} = & \beta_1 \text{Exposure}_{h,t} + \beta_2 \left(\text{Exposure}_{h,t} \times \text{Employment beta}_{c,t-1} \right) \\ & + X'_{h,t-1} \Gamma + \gamma_{c,t} + \varepsilon_{i,c,t}. \end{aligned} \quad (7)$$

The coefficient of interest is β_2 .

Interpretation. If $\beta_2 < 0$, then banks more affected by stress tests reduce lending more in riskier counties. This is the central evidence that stress tests shift lending away from risky markets.

3.3 Bank-year fixed-effects version

Some specifications replace BHC controls with bank-by-year fixed effects:

$$\text{Loan growth}_{i,c,t} = \beta_2 \left(\text{Exposure}_{h,t} \times \text{Employment beta}_{c,t-1} \right) + \alpha_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}. \quad (8)$$

Interpretation. This absorbs all bank-year shocks, including capital, risk appetite, business model, and management changes at that time. Identification then comes from whether the same bank changes lending more in riskier versus safer counties.

3.4 Local versus non-local CRA quantities

The paper estimates the quantity model separately for counties where the bank has a branch and counties where it does not.

Interpretation. If banks can price risk using local information, then they may keep lending in local markets and raise rates. If they lack local information, they may reduce quantities instead. This is why the local/non-local split is central to the mechanism.

3.5 Loan-level pricing model

The STBL pricing regression is

$$\text{Loan rate}_{\ell,i,s,t} = \beta_1 \text{Exposure}_{h,t-1} + Z'_{\ell,i,s,t} \Theta + X'_{h,t-1} \Gamma + \delta_{s,t} + \varepsilon_{\ell,i,s,t}, \quad (9)$$

where ℓ indexes loans and s indexes borrower state.

$Z_{\ell,i,s,t}$ includes loan-level controls such as loan size, risk rating, maturity, collateral, floating-rate status, SBA guarantee, syndication, and prepayment penalty. $\delta_{s,t}$ is a state-by-quarter fixed effect.

Interpretation. This model asks whether loans made by more affected stress-tested banks carry higher interest rates after controlling for borrower risk and local credit conditions.

3.6 Pricing with local branch presence

The local-pricing model is

$$\begin{aligned} \text{Loan rate}_{\ell,i,s,t} = & \beta_1 \text{Exposure}_{h,t-1} + \beta_2 \left(\text{Exposure}_{h,t-1} \times \text{Local}_{i,s,t} \right) \\ & + Z'_{\ell,i,s,t} \Theta + X'_{h,t-1} \Gamma + \delta_{s,t} + \varepsilon_{\ell,i,s,t}. \end{aligned} \quad (10)$$

Interpretation. β_1 captures the pricing effect in non-local markets. $\beta_1 + \beta_2$ captures the pricing effect in local markets. A positive β_2 means stress-tested banks raise prices more where they have local knowledge.

3.7 Pricing by borrower risk

The paper also estimates the local-pricing model separately for low-risk, medium-risk, and high-risk loans.

Interpretation. If stress tests increase the capital cost of risk, then price increases should be concentrated among risky loans, not safe loans.

3.8 Risky-share model

To study the risk composition of lending, the paper estimates

$$\begin{aligned} \text{Risky share}_{i,s,t} = & \beta_1 \text{Exposure}_{h,t-1} + \beta_2 \left(\text{Exposure}_{h,t-1} \times \text{Local}_{i,s,t} \right) \\ & + \phi \text{Average loan size}_{i,s,t} + X'_{h,t-1} \Gamma + \delta_{s,t} + \varepsilon_{i,s,t}. \end{aligned} \quad (11)$$

$\text{Risky share}_{i,s,t}$ is the share of loans in the highest risk category, rating 4.

Interpretation. A negative β_1 means more affected banks reduce risky lending in non-local markets. A positive interaction with local lending means this reduction is weaker or absent where the bank has local information.

3.9 Aggregate county-level credit model

To test whether bank-level contractions reduce total county credit, the paper estimates

$$\begin{aligned} \text{Loan growth}_{c,t} = & \beta_1 \text{Local banks exposure}_{c,t} + \beta_2 \text{Non-local banks exposure}_{c,t} \\ & + W'_{c,t-1} \Pi + \mu_c + \tau_t + \varepsilon_{c,t}. \end{aligned} \quad (12)$$

$W_{c,t-1}$ includes county-level controls such as home-price growth, employment growth, and personal-income growth.

Interpretation. This model asks whether counties exposed to more affected stress-tested banks experience lower aggregate small-business loan growth.

3.10 Small local bank market-share model

The market-share specification is

$$\begin{aligned} \text{Small bank share}_{c,t} = & \beta_1 \text{Local banks exposure}_{c,t} + \beta_2 \text{Non-local banks exposure}_{c,t} \\ & + W'_{c,t-1} \Pi + \mu_c + \tau_t + \varepsilon_{c,t}. \end{aligned} \quad (13)$$

Interpretation. A positive β_2 means small local banks gain market share where stress-tested non-local banks are more affected by stress tests.

4 Identification and credibility (what makes the estimates credible)

4.1 Within-stress-tested-bank design

The paper avoids a simple comparison of stress-tested and non-stress-tested banks. Instead, it studies variation among banks owned by stress-tested BHCs.

Why this matters. Stress-tested banks are much larger than non-tested banks. Comparing them directly would confound regulation with scale, product mix, funding model, geography, and risk appetite.

4.2 Local demand controls

The CRA quantity regressions include county-by-year fixed effects. The STBL pricing regressions include state-by-quarter fixed effects.

Why this matters. These fixed effects absorb local economic shocks and local credit-demand changes. The estimates are therefore not driven simply by a county or state having weak demand for small-business credit.

4.3 Risk-interaction strategy

Rather than relying only on the direct effect of stress-test exposure, the core quantity test interacts exposure with local-market risk.

Why this matters. If more risk-taking banks both have higher stress-test exposure and grow lending faster, the direct effect may be biased upward. The risk-interaction test asks a sharper question: do more exposed banks retreat specifically from riskier markets?

4.4 Branch presence as a mechanism test

The paper separately studies local and non-local markets. Local markets are those where the bank has a branch presence.

Why this matters. The hypothesis is not merely that stress tests reduce lending. It is that banks can *price* risky loans where they have local information, but *exit* or reduce quantities where they do not.

4.5 Timing

For STBL pricing regressions, stress-test results are matched to subsequent quarterly loan surveys.

Why this matters. This reduces the concern that contemporaneous loan pricing causes stress-test outcomes. It also matches the policy timing: banks observe test results and then adjust lending terms.

4.6 Reverse causality

Reverse causality is unlikely because stress-test outcomes are based on the whole BHC portfolio and forward-looking supervisory scenarios. Small-business loans are a small share of large BHC balance sheets.

Interpretation. It is hard for small-business lending in a particular county to mechanically determine the BHC's stress-test exposure.

4.7 Robustness to bank-year and BHC-year heterogeneity

The paper uses bank-by-year fixed effects in CRA quantity specifications and BHC-year fixed effects in pricing specifications.

Why this matters. These specifications absorb broad unobserved shocks at the bank-year or BHC-year level. The remaining evidence comes from differences across risk segments, local versus non-local markets, or borrower risk classes.

4.8 What the paper cannot fully prove

The paper cannot fully determine whether stress tests impose an optimal capital charge on small-business lending or an excessive one. Both interpretations are consistent with banks reducing risky lending and increasing rates on risky loans.

Interpretation. The paper is very strong on the positive question—how banks respond—but more cautious on the normative question—whether the response is welfare improving.

5 Tables and figures

5.1 Figures 1–2: lending trends

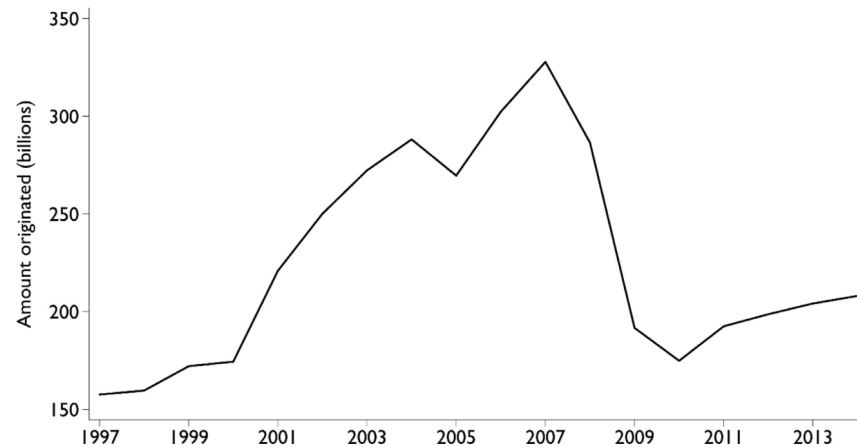


Fig. 1. Total small business lending originations. The figure shows the total volume of small business loans originated by the Community Reinvestment Act (CRA) reporting banks in the USA between 1997 and 2015. The data are aggregated to the national level from the CRA institution-level Disclosure Reports covering the lending activity of all CRA-reporting institutions.

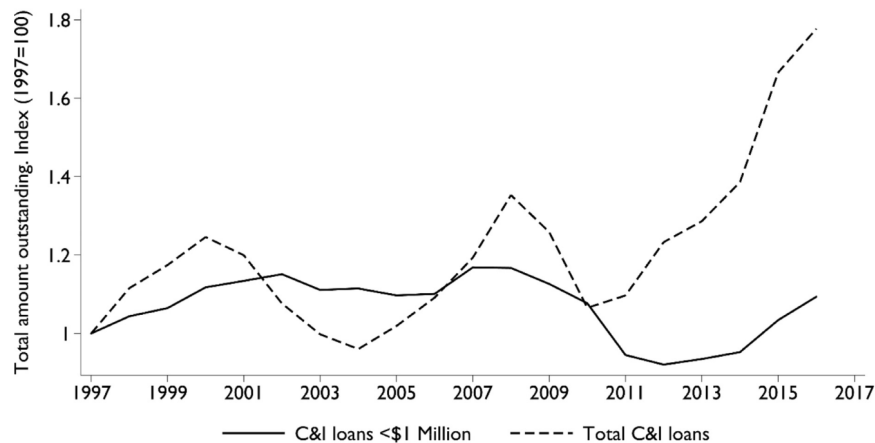


Fig. 2. Total outstanding C&I loans. The figure shows the total amount of outstanding commercial and industrial (C&I) loans split by loan size between 1997 and 2016, indexed to 1997, in 2007 US dollars. The data are from the Consolidated Reports of Condition and Income (Call Reports), reported in June of each year. Non-commercial banks, foreign-controlled banks (with foreign ownership larger than 25%), and banks with missing data for assets, loans, equity, and deposits are excluded.

Figures 1–2: aggregate small-business originations and outstanding C&I loans.

Read:

- Small-business originations rose before the financial crisis and then fell sharply around 2008–2009.
- Total C&I lending recovered strongly after the crisis.
- Small C&I loans recovered more slowly than overall C&I credit.

Economic message. The paper starts from a real policy concern: small-business lending recovered slowly after the crisis. Stress testing is one possible explanation, but the paper asks whether the evidence supports that view.

5.2 Table 1 and Figure 3: summary statistics and stress-test exposure

Table 1

Summary statistics.

This table reports summary statistics for the *Stress-test Exposure* measures, bank characteristics, growth in small business loan originations, and small business lending loan terms. Data sources are public release of the stress-test results by the Federal Reserve, Consolidated Report of Condition and Income (Call Reports), Community Reinvestment Act (CRA) data, and Survey of Terms of Business Lending (STBL), respectively.

	Mean	Median	Standard deviation
<i>Panel A: Stress-test exposure (2012–2016)</i>			
Minimum stress-test distance	2.57	2.70	1.41
Tier 1 capital distance	2.86	2.80	1.31
Tier 1 capital stress-test exposure	3.48	2.90	1.81
<i>Panel B: BHC characteristics (2012–2016)</i>			
Log of bank assets	19.45	19.04	1.15
Deposits/total liabilities	70.9%	78.8%	19.4%
NPL/loans	2.2%	1.9%	1.6%
ROA	0.9%	0.9%	0.6%
C&I loans/assets	12.9%	13.5%	7.7%
Cash + securities/assets	22.3%	23.1%	5.6%
Consumer loans/assets	11.1%	8.1%	14.6%
Mortgage loans/assets	22.6%	23.9%	13.6%
Trading assets/assets	5.2%	1.2%	6.9%
Growth in CRA loans	0.5%	–0.6%	25.2%
<i>Panel C: STBL loan terms (Q2 2012– Q2 2017)</i>			
Loan rate (percentage points)	3.38	3.40	1.27
Log of loan size	11.14	10.97	1.11
Maturity (months)	15.27	12.00	15.94
Rating (1 = safest; 4 = riskiest)	3.17	3.00	0.76
Share of risky loans (Rating = 4)	0.35	0.00	0.48
Local lender? (branch in borrower's state)	61%	–	–
Loan is secured?	88.7%	–	–
SBA loan?	1.7%	–	–
Syndicated loan?	11.9%	–	–
Prepayment penalty?	11.3%	–	–
Floating rate loan?	56.4%	–	–
<i>Panel D: Local-market risk measures</i>			
Employment beta	1.29	1.12	0.69

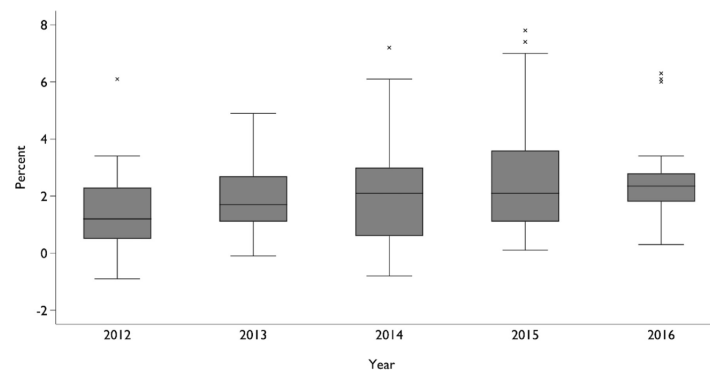


Fig. 3. Distribution of minimum stress-test distance. This figure shows the cross-sectional distribution of Minimum stress-test distance for each year between 2012 and 2016. For each BHC-year, Minimum stress-test distance = $\min(\text{Stressed Tier 1 capital} - 6\%, \text{Stressed total risk-based capital} - 8\%, \text{Stressed leverage ratio} - 4\%)$. An "x" marks an outlier.

Table 1 and Figure 3: key summary statistics and distribution of minimum stress-test distance.

Read:

- Stress-test exposure varies meaningfully across BHC-years.
- The typical BHC has a few percentage points of distance from binding stressed capital thresholds.
- STBL loans average roughly 3.4 percentage points in interest rates and are often secured.
- The county employment beta provides a measurable proxy for local market cyclicity.

Economic message. The identifying variation comes from differences in how strongly the stress tests affect BHCs, not just from whether a bank is formally stress tested.

5.3 Table 2: loan quantities and market risk

Read:

- The interaction between stress-test exposure and county employment beta is negative.
- This result appears across the three exposure measures.
- The result survives bank-by-year fixed effects, which absorb bank-year shocks.
- A one-standard-deviation increase in stress-test exposure implies about a 2.5% greater decline in small-business originations in high-beta counties relative to low-beta counties.

Economic message. Stress tests make affected banks reduce lending more in risky markets. This is a risk-reallocation result, not a simple uniform credit contraction.

Table 2

The effect of stress tests on loan growth by market.

This table reports the results from the (Ordinary Least Squares) regressions following Eq. (2) and Eq. (4). The dependent variable is the growth in small business loan originations by bank i in county c at time t , as reported by the CRA data. An increase in Stress-test exposure implies that a BHC is more affected by its stress-test results. County-year-level Employment beta is calculated as the weighted average of industry betas based on the shares of different industries in a county-year. BHC-level controls are log of total assets, share of C&I loans in a bank total loan portfolio, share of non-performing loans in total loans, return on assets, share of deposits in total liabilities, share of consumer loans, share of mortgage loans, share of trading assets, and a bank's liquidity captured by the ratio of cash and marketable securities to assets. The sample covers the period between 2012 and 2015. Standard errors are clustered by BHC-year. T-statistics are reported in parentheses. "" denotes significance at the 10% level, "" the 5% level, and "" the 1% level.

	Dependent variable = growth in CRA loan originations								
	Panel A: minimum stress-test distance			Panel B: Tier 1 capital distance			Panel C: Tier 1 capital stress-test exposure		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Stress-test exposure	0.106 1.36	0.169** 2.03		0.126* 1.92	0.201*** 3.00		0.110** 2.02	0.149*** 2.65	
Stress-test exposure $\times$ county employment beta		-0.056*** (2.82)	-0.044** (2.53)		-0.067*** (3.19)	-0.057*** (2.85)		-0.035*** (2.11)	-0.026** (2.42)
Initial Tier 1 capital	0.083 (1.26)	0.084 (1.27)		0.086 (1.34)	0.087 (1.36)		- (-)	- (-)	
County $\times$ year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Other BHC controls	Yes	Yes	-	Yes	Yes	-	Yes	Yes	-
Bank $\times$ year fixed effects	-	Yes	-	-	Yes	-	-	Yes	-
Observations	102,563	102,543	102,539	102,563	102,543	102,539	102,563	102,543	102,539
R ²	0.21	0.21	0.57	0.22	0.22	0.57	0.22	0.22	0.57

Table 3

The effect of stress tests on loan growth in local and non-local markets.

This table reports the results from the OLS regressions following Eq. (4). Panel A reports the results for local lenders, those that own a branch in the county. Panel B reports the result for non-local lenders. An increase in Stress-test exposure implies that a BHC is more affected by its stress-test results. County-year-level Employment beta is calculated as the weighted average of industry betas based on the shares of different industries in a county. BHC-level controls are log of total assets, share of C&I loans in a bank total loan portfolio, share of non-performing loans in total loans, return on assets, share of deposits in total liabilities, share of consumer loans, share of mortgage loans, share of trading assets, and a bank's liquidity captured by the ratio of cash and marketable securities to assets. The sample covers the period between 2012 and 2015. Standard errors are clustered by BHC-year. T-statistics are reported in parentheses. "" denotes significance at the 10% level, "" the 5% level, and "" the 1% level.

	Dependent variable = growth in CRA loan originations					
	Minimum stress-test distance		Tier 1 capital distance		Tier 1 capital stress-test exposure	
Panel A: local banks						
Stress-test exposure	-0.089** (2.22)		-0.037 (1.17)		-0.046* (1.89)	
Stress-test exposure $\times$ county employment beta	0.019 (0.81)	0.019 (1.16)	-0.011 (0.47)	0.011 (0.73)	-	0.010 (0.91)
Initial Tier 1 capital	-0.016 (0.67)		-0.019 (0.76)		-	
County $\times$ year effects	Yes	Yes	Yes	Yes	Yes	Yes
Other BHC controls	Yes	-	Yes	-	Yes	-
Bank $\times$ year effects	-	Yes	-	Yes	-	Yes
Observations	16,610	16,587	16,610	16,587	16,610	16,587
R ²	0.34	0.44	0.34	0.44	0.34	0.44
Panel B: non-local banks						
Stress-test exposure	0.222** (2.25)		0.251*** (3.20)		0.183** (2.57)	
Stress-test exposure $\times$ county employment beta	-0.056*** (2.53)	-0.043** (2.33)	-0.072*** (3.31)	-0.055*** (2.64)	-0.041*** (2.24)	-0.030*** (2.88)
Initial Tier 1 capital	0.08 (0.97)		0.086 (1.05)		-	
County $\times$ year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Other BHC controls	Yes	-	Yes	-	Yes	-
Bank $\times$ year fixed effects	-	Yes	-	Yes	-	Yes
Observations	82,891	82,890	82,891	82,890	82,891	82,890
R ²	0.28	0.66	0.29	0.66	0.29	0.66

5.4 Table 3: local versus non-local quantity effects

Read:

- In local markets, the interaction between stress-test exposure and market risk is close to zero.
- In non-local markets, the interaction is negative and significant.
- The pattern is stable across alternative exposure measures.

Economic message. Banks do not mainly cut quantities where they have local branch information. They cut quantities where they are lending from outside the local branch network.

5.5 Table 4: loan pricing

Read:

- Stress-test exposure is positively related to loan interest rates.
- The effect remains after loan controls, BHC controls, and state-quarter fixed effects.
- Larger loans have lower rates and riskier loans have higher rates, as expected.
- A one-standard-deviation increase in exposure implies roughly a 27 basis point increase in rates in the baseline calculation.

Economic message. Stress tests raise the price of small-business credit supplied by affected banks.

Table 4

The effect of stress tests on loan interest rates.

This table reports the results from the loan-level OLS regressions following Eq. (5) with additional time-varying BHC control variables. The dependent variable is loan-level interest rate. The sample covers small business loans originated by the stress-tested banks between 2012Q2 and 2017Q2 and reported in the STBL. An increase in Stress-test exposure implies that a BHC is more affected by its stress-test results. BHC-level controls are similar to those utilized in Table 2. Loan controls are listed in Table 1. Standard errors are clustered by BHC-year. T-statistics are reported in parentheses. “” denotes significance at the 10% level, “” the 5% level, and “” the 1% level.

	Dependent variable = loan interest rate (in percentage points)								
	Panel A: minimum stress-test distance			Panel B: Tier 1 capital distance			Panel C: Tier 1 capital stress-test exposure		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Stress-test exposure	0.193*** (5.52)	0.162*** (4.80)	0.163*** (4.10)	0.202*** (5.33)	0.158*** (4.87)	0.151*** (4.37)	0.194*** (5.07)	0.194*** (5.23)	0.212*** (5.35)
Log loan size	-0.161*** (4.65)	-0.171*** (5.35)	-0.122*** (4.36)	-0.164*** (4.76)	-0.172*** (5.36)	-0.122*** (4.76)	-0.164*** (5.25)	-0.169*** (3.80)	-0.113*** (3.80)
Rating (1 = safest; 4 = riskiest)	0.289*** (4.43)	0.339*** (4.82)	0.425*** (8.13)	0.285*** (4.39)	0.338*** (4.81)	0.427*** (8.21)	0.283*** (4.26)	0.339*** (4.81)	0.425*** (8.01)
Log bank assets	-0.044 (1.01)	-0.012 (0.13)	-0.128 (1.11)	-0.032 (0.78)	-0.042 (0.48)	-0.154 (1.36)	-0.031 (0.76)	-0.04 (0.45)	-0.166 (1.43)
Initial Tier 1 capital ratio	0.148*** (3.58)	0.235*** (4.23)	0.345*** (5.37)	0.185*** (4.01)	0.252*** (4.30)	0.309*** (5.06)	- (4.30)	- (5.06)	- (5.06)
Local lender?	-0.332*** (4.13)	-0.053 (0.68)	-0.022 (0.30)	-0.381*** (4.38)	-0.059 (0.76)	-0.026 (0.34)	-0.347*** (4.37)	-0.069 (0.90)	-0.065 (0.84)
State × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Other BHC controls	-	Yes	Yes	-	Yes	Yes	-	Yes	Yes
Other loan controls	-	-	Yes	-	Yes	Yes	-	Yes	Yes
Observations	498,816	498,816	305,337	498,816	498,816	305,337	498,816	498,816	305,337
R ²	0.25	0.29	0.41	0.25	0.29	0.41	0.25	0.28	0.40

Table 7

The effect of stress tests on loan interest rates: local v. non-local.

This table reports the results from the OLS regressions following Eq. (6). The dependent variable is loan-level interest rate. The sample covers small business loans originated by the stress-tested banks between 2012Q2 and 2017Q2 and reported in the STBL. An increase in Stress-test exposure implies that a BHC is more affected by its stress-test results. Standard errors are clustered by BHC-year. T-statistics are reported in parentheses. “” denotes significance at the 10% level, “” the 5% level, and “” the 1% level.

	Dependent variable = loan interest rate (in percentage points)								
	Panel A: minimum stress-test distance			Panel B: Tier 1 capital distance			Panel C: Tier 1 capital stress-test exposure		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Stress-test exposure	0.102*** (3.10)	0.083*** (2.56)	0.066 (1.38)	0.125*** (3.48)	0.090** (2.54)	0.105** (2.34)	0.103*** (3.56)	0.100*** (3.71)	0.060 (1.47)
Local lender?	0.143 (0.85)	0.413*** (2.55)	0.311* (1.98)	0.053 (0.28)	0.324* (1.86)	0.158 (0.97)	-1.019*** (6.15)	-0.666*** (5.67)	-0.584*** (4.33)
Local Lender × stress-test exposure	0.169*** (3.39)	0.162*** (3.83)	0.113*** (2.73)	0.134*** (2.44)	0.121*** (2.67)	0.057 (1.42)	0.198*** (4.52)	0.193*** (5.05)	0.177*** (4.13)
Log loan size	-0.159*** (4.65)	-0.170*** (5.31)	-0.122*** (4.36)	-0.163*** (4.75)	-0.171*** (5.32)	-0.122*** (4.35)	-0.170*** (5.10)	-0.173*** (5.46)	-0.114*** (3.85)
Rating (1 = safest; 4 = riskiest)	0.284*** (4.36)	0.332*** (4.76)	0.423*** (8.09)	0.282*** (4.33)	0.334*** (4.76)	0.427*** (8.19)	0.280*** (4.30)	0.333*** (4.83)	0.414*** (7.86)
Log bank assets	-0.061 (1.36)	-0.018 (0.20)	-0.128 (1.11)	-0.038 (0.92)	-0.048 (0.55)	-0.15 (1.33)	-0.065 (1.55)	-0.042 (1.55)	-0.191* (0.49)
Initial Tier 1 capital	0.149*** (3.63)	0.235*** (4.37)	0.338*** (5.35)	0.183*** (4.03)	0.248*** (4.36)	0.361*** (5.64)	- (-)	- (-)	- (-)
State × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Other BHC controls	-	Yes	Yes	-	Yes	Yes	-	Yes	Yes
Other Loan controls	-	-	Yes	-	Yes	Yes	-	Yes	Yes
Number of observations	498,816	498,816	305,337	498,816	498,816	305,337	498,816	498,816	305,337
R ²	0.25	0.29	0.41	0.25	0.29	0.41	0.25	0.29	0.41

5.6 Table 7: local versus non-local pricing

Read:

- The interaction between local lending and stress-test exposure is positive in most specifications.
- Stress-test exposure raises rates more where banks have a local branch presence.
- The paper’s baseline calculation implies about a 38 basis point local-market increase versus about 14 basis points in non-local markets.

Economic message. Local information allows affected banks to price the stress-test-induced capital burden instead of exiting.

5.7 Table 8: pricing by borrower risk

Read:

- Stress-test exposure has little reliable effect on low-risk loan rates.
- Medium-risk and high-risk loan rates rise with stress-test exposure.
- The effect is strongest for high-risk loans in local markets.
- For high-risk local loans, a one-standard-deviation increase in exposure implies about a 50 basis point rate increase in the paper’s benchmark calculation.

Economic message. The pricing response is not a blanket rate increase. It is targeted toward the loans that are most costly under stress: risky small-business loans.

5.8 Table 9: risky loan share

Read:

- In non-local markets, stress-test exposure reduces the share of loans in the highest risk category.
- The local-lender interaction is positive, meaning the risky-share reduction is much smaller where the bank is local.
- A one-standard-deviation increase in exposure is associated with roughly a 6–12 percentage point reduction in risky share in non-local markets.

Economic message. The price results reflect a real shift in portfolio risk. Affected banks reduce risky lending where they cannot rely on local information.

Table 8

The effect of stress tests on loan interest rates, by risk rating.

This table reports the results from the OLS analysis following Eq. (5), split by borrower risk rating. The dependent variable is loan-level interest rate. The sample covers small business loans originated by the stress-tested banks between 2010Q2 and 2017Q2 and reported in the STBL. Each panel reports results for different loan risk rating subsamples. These risk ratings are assigned by originating bank and are mapped to a scale provided by the Federal Reserve for better comparison across banks. An increase in Stress-test exposure indicates that a BHC is more affected by its stress-test results. Standard errors are clustered by BHC-year. T-statistics reported in parentheses. ** denotes significance at the 10% level, *** the 5% level, and **** the 1% level.

	Dependent variable = loan interest rate (in percentage points)					
	Minimum stress-test distance		Tier 1 capital distance		Tier 1 capital stress-test exposure	
Panel A: Low risk loans (rating = 1 or 2)						
Stress-test exposure	-0.001 (0.05)	0.023 (1.12)	0.007 (0.25)	0.013 (0.65)	0.016 (0.78)	0.018 (0.91)
Local lender?	-0.552*** (5.72)	0.014 (0.16)	-0.579*** (5.57)	-0.007 (0.08)	-0.361*** (2.27)	0.506*** (3.37)
Local lender × stress-test exposure	0.025 (0.63)	-0.051 (1.57)	0.015 (0.37)	-0.055* (1.80)	-0.072* (1.90)	-0.109*** (3.54)
State × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Control for size	Yes	Yes	Yes	Yes	Yes	Yes
Control for initial Tier 1 capital	Yes	Yes	Yes	Yes	Yes	Yes
Other loan and BHC controls	-	Yes	-	Yes	-	Yes
Number of observations	84,576	84,576	84,576	84,576	84,576	84,576
R ²	0.40	0.46	0.40	0.46	0.40	0.46
Panel B: medium risk loans (Rating = 3)						
Stress-test exposure	0.055* (1.80)	0.034 (1.33)	0.063* (1.84)	0.029 (1.02)	0.078*** (3.33)	0.069*** (2.82)
Local lender?	0.245 (1.60)	0.233 (1.59)	0.244 (1.31)	0.239 (1.49)	-0.394*** (5.55)	-0.723*** (5.89)
Local lender × stress-test exposure	0.228*** (5.02)	0.178*** (4.74)	0.214*** (4.06)	0.162*** (3.91)	0.177*** (4.11)	0.152*** (4.64)
State × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Control for size	Yes	Yes	Yes	Yes	Yes	Yes
Control for initial Tier 1 capital	Yes	Yes	Yes	Yes	Yes	Yes
Other loan and BHC controls	-	Yes	-	Yes	-	Yes
Number of observations	231,115	231,115	231,115	231,115	231,115	231,115
R ²	0.34	0.41	0.34	0.41	0.34	0.41
Panel C: High risk loans (rating = 4)						
Stress-test exposure	0.176*** (4.67)	0.055 (1.32)	0.198*** (5.11)	0.073 (1.65)	0.198*** (6.48)	0.114*** (3.30)
Local lender?	0.489*** (5.52)	0.439*** (2.36)	0.322 (1.57)	0.339* (1.75)	-0.740*** (3.91)	-0.501*** (3.77)
Local lender × stress-test exposure	0.180*** (3.06)	0.128*** (2.65)	0.124*** (2.09)	0.094*** (2.09)	0.189*** (3.78)	0.172*** (3.28)
State × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Control for size	Yes	Yes	Yes	Yes	Yes	Yes
Control for initial Tier 1 capital	Yes	Yes	Yes	Yes	Yes	Yes
Other loan and BHC controls	-	Yes	-	Yes	-	Yes
Number of observations	183,125	183,125	183,125	183,125	183,125	183,125
R ²	0.21	0.27	0.21	0.28	0.21	0.27

5.9 Table 11: alternative local-risk measures

Read:

- The paper repeats the quantity analysis using GDP beta, STBL risky share, low-credit-score share, and small-establishment employment beta.
- The broad pattern remains: stress-test-affected banks reduce lending in riskier markets, especially non-local markets.
- Some alternative measures are noisier, but the direction supports the main employment-beta result.

Economic message. The baseline result is not purely an artifact of one risk proxy.

Table 11

Robustness tests using alternative local market risk measures.

This table reports the results from the OLS regressions similar to those reported in Tables 2 (full sample) and 3 (local vs. non-local banks), with alternative measures of local market risk. (See Table A3 for definitions.) The dependent variable is the growth in small business loan originations by bank i in county c at time t , as reported by the CRA data. We classify a county as a bank's local market if the bank has a branch in that county. The sample covers the period between 2012 and 2015. The Stress-test exposure measure is based on the (negative) Minimum stress-test distance measure for stress-test exposure. Specification (1), (3), and (5) include county-year fixed effects and BHC controls. Specifications (2), (4), and (6) include county-year fixed effects and BHC-year fixed effects. Standard errors are clustered by BHC-year. T-statistics are reported in parentheses. ** denotes significance at the 10% level, *** the 5% level, and **** the 1% level.

	Dependent variable = growth in CRA loan originations					
	All banks (1)	Local banks (2)	Local banks (3)	Non-local banks (4)	Non-local banks (5)	Non-local banks (6)
Panel A: risk measure based on GDP beta						
Stress-test exposure	0.151* (1.97)	-	-0.087* (1.91)	-	0.209** (2.28)	-
Stress-test exposure × county risk measure	-0.040** (2.29)	-0.034* (1.81)	0.015 (0.71)	0.003 (0.23)	-0.040** (2.15)	-0.035* (1.84)
Initial Tier 1 capital	0.075 (1.11)	-	-0.013 (0.54)	-	0.076 (0.88)	-
Observations	102,543	102,539	16,610	16,587	82,891	82,890
R ²	0.21	0.57	0.34	0.44	0.28	0.66
Panel B: risk measure based on lag of STBL share of high-risk loans (risk rating = 4 at BHC-year-state level)						
Stress-test exposure	-0.078 (0.79)	-	-0.039 (1.23)	-	-0.08 (0.92)	-
Stress-test exposure × BHC risk measure	-0.162*** (3.97)	-0.0400* (1.72)	-0.019 (1.01)	0.002 (0.09)	-0.151*** (3.35)	-0.026 (1.62)
Initial Tier 1 capital	0.102** (2.18)	-	-0.014 (0.82)	-	0.159*** (2.86)	-
Observations	36,519	36,517	9,890	9,883	25,530	25,527
R ²	0.37	0.49	0.37	0.41	0.51	0.62
Panel C: risk measure based on share of consumers with credit score < 620						
Stress-test exposure	0.142 (1.63)	-	-0.096* (2.25)	-	0.204* (1.94)	-
Stress-test exposure × county risk measure	-0.169 (1.13)	-0.156** (2.12)	0.098 (1.35)	-0.059 (1.18)	-0.198 (1.11)	-0.112 (1.62)
Initial Tier 1 capital	0.077 (1.14)	-	-0.013 (0.53)	-	0.079 (0.91)	-
Observations	102,283	102,279	16,610	16,587	82,635	82,634
R ²	0.21	0.57	0.34	0.44	0.28	0.66
Panel D: risk measure based on beta of employment in small establishments (< 50 employees)						
Stress-test exposure	0.175** (2.31)	-	-0.103 (1.46)	-	0.226** (2.42)	-
Stress-test exposure × county risk measure	-0.179*** (2.93)	-0.123** (2.36)	0.075 (0.64)	-0.048 (0.63)	-0.167*** (2.52)	-0.107** (2.09)
Initial Tier 1 capital	0.075 (1.11)	-	-0.013 (0.54)	-	0.075 (0.88)	-

Table 9

The effect of stress tests on the share of high-risk loans.

This table reports the results from the OLS regressions following Eq. (7). The dependent variable is Risky share, which is defined as the share of high-risk (rating = 4) loans originated by bank i in state s during year t , as reported in the STBL. The sample covers small business loans originated by the stress-tested banks between 2010Q2 and 2017Q2 and reported in the STBL. An increase in Stress-test exposure implies that a BHC is more affected by its stress-test results. Standard errors are clustered by BHC-year. T-statistics reported in parentheses. ** denotes significance at the 10% level, *** the 5% level, and **** the 1% level.

	Dependent variable = risky share, the share of loans in risk category 4					
	Panel A: minimum stress-test distance	Panel B: Tier 1 capital distance		Panel C: Tier 1 capital stress-test exposure		
Stress-test exposure	-0.078*** (3.28)	-0.044** (2.56)	-0.069*** (4.24)	-0.050*** (3.63)	-0.050*** (4.70)	-0.054*** (4.14)
Local lender?	0.142*** (3.38)	0.105*** (2.36)	0.105*** (2.52)	0.082*** (2.04)	-0.136*** (3.27)	-0.187*** (4.35)
Local lender × stress-test exposure	0.059*** (3.87)	0.059*** (3.86)	0.040*** (3.52)	0.040*** (3.66)	0.034*** (3.32)	0.045*** (3.80)
Log assets	0.049** (2.51)	0.024 (0.75)	0.051*** (2.38)	0.030 (0.92)	0.050** (2.32)	0.025 (0.79)
Initial Tier 1 capital	-0.011 (1.17)	-0.024* (1.83)	-0.036*** (3.49)	-0.044*** (3.12)	-	-
Average loan size	-0.051*** (4.20)	-0.040*** (4.40)	-0.052*** (3.86)	-0.040*** (4.21)	-0.056*** (3.91)	-0.049*** (3.54)
State × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Other BHC controls	Yes	Yes	-	Yes	Yes	Yes
Observations	10,216	10,216	10,216	10,216	10,216	10,216
R ²	0.16	0.28	0.17	0.29	0.16	0.29

5.10 Table 12: aggregate credit and substitution by small banks

Read:

- Panel A finds little evidence that county-level small-business lending growth falls when counties are exposed to affected stress-tested banks.
- Panel B shows that small local banks increase market share where non-local stress-tested banks are more affected.
- The substitution effect is strongest for exposure to non-local stress-tested lenders.

Economic message. Stress tests change the composition of credit supply. They do not necessarily reduce total credit because smaller local banks partially replace large non-local stress-tested banks.

6 Short summary

- The paper studies whether Federal Reserve stress tests affect small-business lending.
- Stress-test exposure is measured by how close stressed capital ratios come to regulatory minimums or how much capital falls under stress.
- In CRA data, more exposed banks reduce lending more in riskier counties.
- This quantity reduction occurs mainly in non-local markets where banks lack a branch presence.
- In STBL data, more exposed banks raise loan interest rates.
- Price increases are concentrated among risky loans and local markets.
- The share of high-risk loans falls in non-local markets for more exposed banks.
- Aggregate county-level credit does not fall, because smaller local banks appear to expand where large stress-tested non-local banks retreat.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

Stress tests affect small-business credit supply. More affected stress-tested banks reduce loan quantities in risky, non-local markets and raise interest rates on risky local loans. The evidence points to risk repricing and risk reallocation rather than a simple across-the-board lending contraction.

7.2 Economic intuition

Stress tests make future capital requirements more sensitive to portfolio losses under a severe downturn. Risky small-business loans become more costly from the bank's perspective. When the bank has local information, it can charge for this risk. When it lacks local information, it is more likely to reduce lending.

7.3 Takeaway

The paper's central message is that stress tests reshape the structure of small-business lending. Large affected banks move away from risky non-local lending, while small local banks partly fill

the resulting gap. The welfare implication depends on whether stress tests correctly price risk or overburden small-business credit, but the positive empirical result is clear: stress tests alter credit allocation through both quantities and prices.